📘 MODULE 4: INDEXES

4.1 Concept

An **index** is a storage location on the indexer. Events go into an index when they're ingested. Searches read from indexes.

4.2 Why more than one index?

1. **Access control:** the HR team can see hr, the SOC sees security.
2. **Retention:** keep firewall logs 1 year, debug logs 7 days.
3. **Performance:** searching index=windows is faster than scanning everything.
4. **Organization:** one index per data type is common.

4.3 Default indexes

Index	Purpose
main	Default index for data when none is specified
_internal	Splunk's own logs (very useful for troubleshooting)
_audit	Who did what in Splunk (searches, logins)
_introspection	Performance data about Splunk
summary	Default for summary-indexed results

Names starting with _ are internal and hidden from normal index=* searches. Use index=_* to see them.

4.4 Creating an index

UI: Settings → Indexes → **New Index** → name (lowercase, e.g. linux_auth) → Save.

Config file (/opt/splunk/etc/system/local/indexes.conf):

ini

```
[linux_auth]
homePath   = $SPLUNK_DB/linux_auth/db
coldPath   = $SPLUNK_DB/linux_auth/colddb
thawedPath = $SPLUNK_DB/linux_auth/thaweddb
frozenTimePeriodInSecs = 7776000
```

- [linux_auth]: stanza = the index name.

- homePath: hot and warm buckets. coldPath: cold buckets. thawedPath: restored data.
- frozenTimePeriodInSecs: retention. 7776000 seconds = 90 days. Older data is frozen (deleted by default).
- $SPLUNK_DB expands to /opt/splunk/var/lib/splunk.
- Restart Splunk to apply.

Mistake: sending data to an index that does not exist. The data is dropped, and the error shows in splunkd.log.

4.5 Buckets: the life of data

HOT → WARM → COLD → FROZEN (deleted/archived) → THAWED (restored)

Stage	Meaning
Hot	Currently being written to. Newest data.
Warm	Closed (rolled from hot), still fast storage.
Cold	Older, often on cheaper storage.
Frozen	Past retention. Deleted unless you configured an archive.
Thawed	Archived data you manually restored for searching.

Data moves along based on bucket size, count, and age. The retention settings are frozenTimePeriodInSecs (age) and maxTotalDataSizeMB (size). Whichever limit is hit first causes the oldest data to freeze.

4.6 Practical searches

Query: which indexes have data?

spl

| eventcount summarize=false index=* | dedup index | fields index

What it does: lists indexes you can access that contain events.
Line-by-line: eventcount counts events per index · summarize=false gives per-server rows · dedup index removes duplicates · fields index keeps one column.
Common mistake: forgetting the leading | because this is a generating command.
Interview explanation: "I use eventcount to see all searchable indexes, or tstats for speed."

Query: where are my events stored?

spl

index=* | stats count by index, sourcetype, host

What it does: shows, for each index, which sourcetypes and hosts it holds.
Mistake: index=* does not include _internal, and it is slow over All time.

4.7 Why index=main works but index=web returns nothing

Cause	Check
The index web was never created	Settings → Indexes
Index exists but no data was sent to it	Input points to a different index
Wrong time range	Set All time
Your role can't search it	Role's "searchable indexes"
Typo or wrong case in the name	Compare to Settings → Indexes

index=main works because it exists, has data, and is usually searchable by default.

4.8 Index sizing (concept)

Estimate: daily volume × retention days × about 50% (Splunk compresses roughly to half, plus index files). Example: 10 GB/day × 90 days × 0.5 ≈ 450 GB. This is a rough rule. Verify against current Splunk sizing docs.

4.9 Hands-on

1. Create an index named linux_auth in the UI.
2. Edit your auth.log monitor so its index is linux_auth.
3. Search index=linux_auth | head 5.
4. Run both queries from 4.6 and report what you see.

4.10 Interview Q&A

- **What is an index?** A repository where Splunk stores events, with its own retention and access control.
- **Why multiple indexes?** Access control, retention, performance, organization.
- **Bucket lifecycle?** Hot → warm → cold → frozen → (thawed).
- **Which settings control retention?** frozenTimePeriodInSecs and maxTotalDataSizeMB in indexes.conf.

- **index=web returns nothing, why?** Index missing, no data, time range, permissions, typo.

4.11 Revision

Index = storage + access + retention · default: main, _internal, _audit · buckets hot/warm/cold/frozen/thawed · config in indexes.conf.